88. Domestic electrical wiring is basically a:

(a) Series connection
(b) Parallel connection
(c) Combination of series and parallel connection
(d) Series connection within each room and parallel connection elsewhere

Ans. (b): In a parallel circuit, the voltage across each of the components is the same, and the total current is the sum of currents through each component. The wiring for most homes is parallel.

89. A simple machine helps a person in doing:

(a) less work
(b) the same amount of work with lesser force
(c) the same amount of work slowly
(d) the same amount of work much faster.

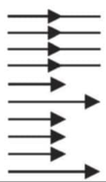
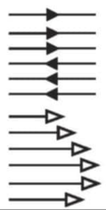
Ans. (b): A simple machine is advice that changes the direction or magnitude of a force. In general, they can be defined as the simplest mechanisms that use mechanical advantage to multiply force. Thus, simple machine helps us in doing same amount of work with lesser force. Few examples of simple machines are pulkey, lever, wheel, screw, etc.

90. When an air bubble at the bottom of lake rises to the top it will

(a) increase in size (b) decrease in size
(c) maintain its size (d) flatten into disc like shape

Ans. (a): When air bubble is at bottom of lake there will be more atmospheric pressure on it and when it will come at the surface of water, there will be lower pressure on it and thus it will be increased in size.

91. A liquid is flowing in an streamlined manner through a cylindrical pipe. Along a section containing the axis of the pipe, the flow profile will be:

(a) 
(b) 
(c)
(d)

Ans. (a): If a fluid flows such that its velocity at a point is always the same in magnitude and direction, the fluid is said to have a streamline flow. The type of flow in a fluid system is characterized by the presence of laminae, or parallel stream of fluid. A streamline is a straight or curved path such that tangent to it and point gives the direction of flow of liquid at that point. Thus, according to question the flow profile is best represented by figure (a).

CHEMISTRY

92. The alpha particle carries two positive charge. Its mass is very nearly equal to that of

(a) two protons
(b) an atom of helium
(c) sum of masses of two positrons and two neutrons
(d) Two positrons as can positron carries a single positive charge

Ans. (b): Each Alpha particle contains two protons and two neutrons.
Total number of nucleons = Number of protons + Number of neutrons
= 2 + 2 = 4
Total number of nucleons = mass number of an element
Thus mass number of alpha particle is 4 which is equal to molecular weight of helium atom

93. Which one of the following statements is alloyed with iron to produce steel which can resist high temperatures and also have high hardness and abrasion resistance?

(a) Aluminum (b) Chromium
(c) Nickel (d) Tungsten

Ans. (b): Steel is an alloy of iron and carbon on mixing with carbon its strength and toughness got increased. In order to make it further more temperature and abrasion resistant chromium is generally mixed with iron. It also increases its hardness and load bearing capacity. Stainless steel is a category of steel consists (75%) Fe 18% Cr and and Ni is corrosion resistant.

94. Living organism require at least 27 elements, of which 15 are metals. Among these, those required in major quantities include.

(a) Potassium, Manganese, Molybdenum and calcium
(b) Potassium, Molybdenum, Copper and calcium
(c) Potassium, Sodium, Magnesium, and calcium
(d) Sodium, Magnesium, Copper and manganese

Ans. (c): Living organism require about 27 elements of which 15 are metals K, Mg, Na, and Ca are required in major. Minor quantity of Mn, Fe, Co, Cu, Zn and Mo and trade amount of Cr, Sn, Ni and Al are required by some organism.
Calcium, magnesium and phosphorus are essential parts of the bones and teeth.
Calcium is necessary for blood clotting sodium helps in metal activity and transmission of nerve.

BIOLOGY

95. Which one of the following sets of conditions is necessary for a good cultivation of Wheat?

(a) Moderate temperature and moderate rainfall (b) High temperature and heavy rainfall
(c) High temperature and moderate rainfall
(d) Low temperature and low rainfall

Ans. (a): Wheat is a temperate crop. It can be grown well in sub-tropical regions. It requires moderate temperature of- 10^0C to 15^0C and moderate rainfall of 50 to 75 cm.

96. Physico chemical characteristics of water in water sources undergo changes due to

(a) aquatic macrophysics
(b) aquatic fungi
(c) effluents
(d) evapotranspiration

Ans. (c): Aquatic macrophytes, aquatic fungi are natural organism and evapotranspiration is a natural process by which plant losses water. Thus being natural they do not result into any change in physico-chemical characteristics of water while effluents discharges into water bodies by industrial units and domestic sewage results into change in physic-chemical characteristics like pH, conductivity, temperatures electrical conductivity, dissolved oxygen Biological oxygen demand, chemical oxygen demand, NH_4^+ and NH_3^- ion contents etc.

97. It is possible to produce seedless tomato fruits by

(a) applying trace elements in tomato fruits
(b) spraying mineral solution on plants
(c) spraying hormones on Flowers
(d) applying fertilizers containing radioactive elements.

Ans. (c): Seedless tomato fruit can be grown without fertilization if the plant growth hormone is sprayed on them. This will result in their parthenocarpy. This is the technique of induced parthenocarpy where the vegetables are sprayed with plant growth

98. Ecologist have so far regulators.
Show the correct option is 'Spraying hormones on flowers'.
known, found and identified a large number
of species in the plant and animal
kingdom. In terms of numbers, the largest
found and identified so far is from among the:

(a) Fungi (b) Plants
(c) Insects (d) Bacteria

Ans. (c): In plant and animal kingdom largest number of species identified are insects.

99. Which of the following are associated with Diabetes mellitus, a common disease in adults?

1. Higher sugar level in blood
2. Lower sugar level in blood
3. Lower insulin level in blood
4. Higher insulin level in blood

Select the correct answer by using the codes given below:

(a) 2 and 4 (b) 1 and 2
(c) 2 and 3 (d) 1 and 3

Ans. (d): Diabetes Mellitus is called due to deficiency of insulin hormone. The hormone insulin maintains the blood sugar in body. Deficiency of insulin in human body causes increase in sugar level in blood, which leads to diabetes mellitus.

100. Consider the following statements AIDS – is transmitted

1. By sexual intercourse
2. By blood transfusion
3. By mosquitoes and other blood sucking insects
4. Across to placenta

(a) 1, 2 and 3 correct (b) 1, 2 and 4 correct
(c) 1, 3 and 4 correct (d) 1 and 3 are correct

Ans. (b): AIDS is a viral disease. The virus is called HIV. This disease is transmitted by sexual intercourse, blood transfusion and from pregnant mother to baby through placenta.

101. Which of the following would lead to malnutrition?

1. Over nutrition
2. Under nutrition
3. Imbalanced nutrition

Select the correct answer by using the codes given below.
Codes:

(a) 2 alone (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

Ans. (d): Malnutrition is defined as the insufficient, excessive or imbalanced consumption of nutrients. So under nutrition, over nutrition and imbalanced nutrition leads to malnutrition.

102. People drinking water from a shallow hand pumps, are likely to suffer from all of the following disease except.

(a) Cholera (b) Typhoid
(c) Jaundice (d) Fluorosis

Ans. (d): Cholera, Typhoid and Jaundice are waterborne disease. In shallow hand pumps there is possibility of microbial contamination (bacteria) to water. But fluorosis occurs due to excess amount of fluorine found in water.